SAFE-ALPHA: Anytime-Valid Certification for Adaptive Strategy Search

Extended technical report with proofs; not part of the ICAIF submission

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Abstract

Adaptive strategy search creates an online multiple-testing problem: proposal identities depend on earlier evaluations, monitoring horizons differ, and the researcher may stop when evidence appears favorable. SAFE-ALPHA separates an arbitrary adaptive proposer from an executable ledger that fixes each strategy, cost rule, bounded score, rank weight, and predictable e-process before its first tested payoff. Joint promotion and proposal-level evidence freezing produce an irreversible set satisfying $\mathbb{E}[\sup_t \mathrm{FDP}(D_t)] \leq q$ under arbitrary serial and cross-strategy dependence. Online e-BH is the testing engine; the contribution is the financial deployment contract and its causal evaluation. In 500 market-shaped sign-null replays at $q = 10\%$, false promotion occurs in 20 runs, while same-bar betting and repeatedly inspected nominal tests fail in every run. At planted annualized Sharpe 1.5 and correlation 0.5, the original persistent rule attains power 0.578, versus 0.479 for a matched terminal rule. A development-frozen finite-campaign prior, daily inspection, and prespecified three-bet mixture reach 0.693 on a new paired holdout, versus 0.572 for the original schedule. A locked U.S. industry replay certifies the market control but none of 76 adaptive symbolic rules; factor residualization removes the control certificate. The evidence concerns the committed bounded score, not raw-return alpha, drawdown, or profitability.

CCS Concepts

• **Applied computing** → **Economics**; • **Mathematics of computing** → **Probability and statistics**; • **Computing methodologies** → *Artificial intelligence*.

Keywords

adaptive strategy search, e-processes, false discovery rate, factor mining, anytime-valid inference, backtest overfitting

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1 Introduction

Consider a research agent that proposes a momentum rule, reads its backtest, changes the lookback, compares the revision with other candidates, and continues until something appears convincing. A held-out period ceases to be held out when its outcome shapes the next proposal, and a conventional test loses its nominal interpretation when it is checked until rejection. No malicious data mining is required; multiplicity and optional stopping arise from the architecture of the research process itself.

Recent systems use language models to propose, implement, debate, retrieve, and evolve financial signals [3, 6, 9, 11, 12]. Several enforce out-of-sample isolation or high terminal test-statistic hurdles. Those safeguards do not by themselves control error across an adaptively growing stream that is inspected and stopped from the complete research record.

SAFE-ALPHA assigns proposal and promotion to different components. The proposer may use the complete recorded past and may itself be stochastic, language-model based, or adversarially persistent. The ledger fixes the candidate's formula, universe, execution lag, missing-data policy, cost model, payoff transformation, and evidence rule before another payoff is observed. The candidate then begins with unit evidence, and every material revision receives a new identity. A non-generative gate alone may add the strategy to the certified set.

The protocol indexes both proposal time and market time. Candidate j arrives at the adaptive research time τ_j, whereas its evidence evolves over later market dates. A terminal e-BH analysis presumes a fixed decision horizon; preserving every interim peak is also invalid because the running maximum of an e-process need not be an e-value [13]. SAFE-ALPHA instead allocates a summable budget when the proposal arrives, applies the joint rule only to arrived candidates, and freezes one e-valid value at the first joint promotion time. The resulting set is irreversible, which permits immediate deployment, yet its expected worst realized false-discovery proportion remains bounded throughout the research run.

SAFE-ALPHA makes three contributions distinct from the underlying e-BH rule. First, it defines an executable financial-research contract in which proposal identities and times may depend on the complete past, while the formula, execution lag, costs, score, rank weight, and evidence rule are fixed before the first tested payoff. Second, it proves that joint promotion followed by proposal-level freezing yields an irreversible set with $\mathbb{E}[\sup_t \mathrm{FDP}(D_t)] \leq q$ under arbitrary serial and cross-strategy dependence, and it gives pathwise power results for persistence and denser inspection. Third, it evaluates the complete causal protocol with valid matched comparators, controlled timing violations, paired power and delay experiments, and a public replay that separates the prespecified control from adaptively selected rules. Online e-BH, stopped e-values, and worst-time FDR control are established statistical ingredients [4, 15]. The contribution here is the deployment contract that makes those ingredients executable in adaptive financial search.

2 Relation to prior work

Multiple testing has long been central to empirical finance. The Reality Check and SPA address the best model in a fixed candidate family [7, 18]; the Deflated Sharpe Ratio and PBO are retrospective diagnostics for selection and backtest overfitting [1, 2]; and the factor-zoo literature documents why ordinary significance thresholds are too permissive [8]. These methods remain useful, but they target a fixed or retrospectively enumerated search rather than hypotheses whose identities, arrival times, and monitoring horizons depend on the evolving research record.

Our gate builds on e-values [14]. e-BH controls FDR for arbitrarily dependent e-values and includes a sequential-trader example [16]; e-value procedures extend to online hypothesis arrival [4, 20]; and stopped e-BH identifies the global filtration required when evolving evidence streams are monitored jointly [15]. The 2026 online e-closure and donation framework strictly improves baseline online e-BH on streams of fixed e-values [19]. It does not directly supply the jointly stopped ledger used here, in which every proposal carries an evolving global e-process, an individual admission floor, and its own frozen promotion value. We therefore use the baseline whose stopped values can be audited proposal by proposal. Our claim concerns neither e-BH nor worst-time FDR in isolation; it concerns their deployment-safe composition with financial payoffs and the empirical evaluation of the causal failures created by adaptive strategy search.

The need for a proposal-time budget can be seen without any finance-specific distributional assumptions. The following counterexample also explains why running terminal e-BH after the agent decides to stop is not a valid repair.

Proposition 1 (A random terminal family can have FDR one). *Let every hypothesis be null and, independently for $j \geq 1$, set*

$$E_j = \begin{cases} j/q, & \text{with probability } q/j, \\ 0, & \text{otherwise.} \end{cases}$$

Every E_j is an e-value. If proposal generation stops at $J = \inf\{j : E_j = j/q\}$, then $J < \infty$ almost surely and ordinary unweighted e-BH applied to $E_1, \ldots, E_J$ rejects with FDR one.

Proof. Each expectation equals one, but independence and $\sum_j q/j = \infty$ imply that a success eventually occurs almost surely. At that success, the one-rejection e-BH boundary for a family of size J is $J/q = E_J$. Since all hypotheses are null, the resulting nonempty discovery set has false-discovery proportion one. $\square$

This construction does not invalidate terminal e-BH for a genuinely fixed family and horizon. It shows that the family size cannot be chosen by the same evidence later submitted to the test, which is precisely what an open-ended proposer would otherwise be allowed to do.

Table 1: The certification contract. Every row is enforced by the ledger rather than by the proposer.

Stage	Committed or permitted information
Proposal	Formula, universe, lag, costs, score, code hash; global past only
Execution	Point-in-time inputs through $t-1$; realized payoff at t
Evidence	Post-proposal scores; predictable bet and scale only
Promotion	Joint weighted gate; evidence frozen on admission
Revision	New identifier, proposal time, weight, and unit evidence

3 The SAFE-ALPHA protocol

3.1 One filtration and an immutable ledger

Let $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\geq 0}, P)$ be a discrete-time filtered probability space. The global filtration contains all market observations, agent inputs and outputs, candidate specifications, evidence streams, and gate decisions available by t. Candidate j is proposed at a global stopping time τ_j. Its complete executable specification A_j is $\mathcal{F}_{\tau_j}$-measurable, and only finitely many candidates have arrived by any finite t. Later proposals may depend on all earlier outcomes. A revision of A_j is a new candidate, not a continuation of j.

For $t > \tau_j$, let $X_{j,t} \in [-1, 1]$ be the one-period net performance score generated by the committed execution and cost rules. The null is

$$H_j : \quad \mathbb{E}_P[X_{j,t} \mid \mathcal{F}_{t-1}] \leq 0 \quad \text{for every } t > \tau_j. \tag{1}$$

This is a conditional no-positive-drift null for the *bounded score*. It is stronger than nonpositive unconditional mean, nonpositive Sharpe, or a zero factor-model intercept. If returns are clipped to construct X, the certificate concerns the clipped score; it must not be described as a certificate for the raw mean return.

Let $\lambda_{j,t} \in [0, 1]$ be $\mathcal{F}_{t-1}$-measurable and define

$$E_{j,t} = \begin{cases} 1, & t \leq \tau_j, \\ \displaystyle\prod_{s=\tau_j+1}^{t} (1 + \lambda_{j,s} X_{j,s}), & t > \tau_j. \end{cases} \tag{2}$$

Under H_j, E_j is a nonnegative supermartingale in the *global* filtration. The distinction is essential. A process valid only in its own local history can cease to be valid when another strategy stream reveals information used to stop it [15]. Operationally, a signal using date-t closes trades no earlier than the next implementable price; data vintages, proposal timestamps, code hashes, random seeds, and cost inputs are committed to the ledger before the corresponding outcome is observed; Table 1 records these fields.

The predictability requirement has observable force. Under a symmetric null $X_t \in \{-1, 1\}$, choosing $\lambda_t = \frac{1}{2}\mathbf{1}\{X_t = 1\}$ after seeing X_t gives expected one-step multiplier 1.25, not at most one. This is the same-bar leakage implemented in our ablation. Likewise, retaining $\sup_{s\le t} E_{j,s}$ generally spends evidence more than once. By contrast, the first joint-promotion time below is a global stopping time, so its single frozen value remains e-valid by optional sampling.

3.2 Summable budget and irreversible gate

Proposal j receives a nonnegative weight $\gamma_j \in \mathcal{F}_{\tau_j}$ with $\sum_{j\ge 1}\gamma_j \le 1$ almost surely. Our default,

$$\gamma_j = \{j(j+1)\}^{-1}, \tag{3}$$

supports an unbounded stream. A data-dependent terminal family size cannot safely replace this proposal-time budget. We use the convention $(q\gamma_j k)^{-1} = \infty$ when $\gamma_j = 0$, and impose the ledger-fixed new-admission floor $h = 1/q$. The floor is not needed for FDR control, but it ensures that every promoted strategy has standalone evidence of at least $1/q$; without it, a weak-evidence candidate can become a passenger after a large set of strong discoveries has lowered the joint threshold.

Let D_{t-1} be the finite set promoted before inspection t. The finite adapted eligibility set obeys

$$D_{t-1} \subseteq \mathcal{A}_t \subseteq \{j : \tau_j < t\}, \tag{4}$$

which formally prevents an unproposed candidate, still carrying unit evidence, from entering the gate. Evidence for a promoted candidate is frozen at its promotion time

$$S_j = \inf\{t \ge \tau_j : j \in D_t\}, \qquad \widetilde{E}_j = E_{j,S_j}.$$

For $j \in \mathcal{A}_t$, form

$$Z_{j,t} = \begin{cases} \widetilde{E}_j, & j \in D_{t-1}, \\ E_{j,t}, & j \in \mathcal{A}_t \setminus D_{t-1}. \end{cases}$$

For a proposed set size k, define

$$\mathcal{C}_t(k) = \left\{ j \in \mathcal{A}_t : Z_{j,t} \ge \max\left(h, \frac{1}{q\gamma_j k}\right) \right\}.$$

The maximal feasible size and the new discovery set are

$$k_t^\star = \max\left(\{0\} \cup \{1 \le k \le |\mathcal{A}_t| : |\mathcal{C}_t(k)| \ge k\}\right), \tag{5}$$

$$D_t = \mathcal{C}_t(k_t^\star), \tag{6}$$

where $\mathcal{C}_t(0) = \varnothing$ and newly admitted evidence is then frozen. If $d = |D_{t-1}|$, the induction invariant makes $k = d$ feasible. Hence $k_t^\star \ge d$, the joint threshold can only fall, and no promotion is withdrawn. If $|\mathcal{C}_t(k_t^\star)| > k_t^\star$, the same set makes the next integer feasible, contradicting maximality; thus $|D_t| = k_t^\star$. Every old member satisfied its threshold when admitted, and every new member satisfies the current one, which yields the invariant

$$j \in D_t \quad \Longrightarrow \quad \widetilde{E}_j \ge \frac{1}{q\gamma_j |D_t|}. \tag{7}$$

as well as $\widetilde{E}_j \ge h$ for every member. Freezing at S_j is not the same as retaining $\sup_{s\le t} E_{j,s}$: only evidence that passes the joint gate at a declared inspection is remembered.

4 Simultaneous error control

Because the proposal itself is random, write N_j for the indicator that the selected candidate is a true null under P. We require N_j to be $\mathcal{F}_{\tau_j}$-measurable and, equivalently to Eq. (1) on the null event,

$$N_j \mathbf{1}\{\tau_j < t\}\mathbb{E}_P[X_{j,t} \mid \mathcal{F}_{t-1}] \le 0.$$

This selective formulation permits the agent to choose A_j from the global past. It prohibits selecting A_j with the return intended to test it.

Theorem 2 (Persistent SAFE-ALPHA certificate). *Suppose τ_j is a global stopping time; A_j, N_j, γ_j are $\mathcal{F}_{\tau_j}$-measurable; finitely many candidates arrive by each finite t; Eq. (1) holds on $\{N_j = 1\}$; $\lambda_{j,t}$ is predictable; $\sum_j \gamma_j \le 1$ almost surely; and the finite adapted gate obeys Eq. (4), is nested, and maintains Eq. (7). Without independence assumptions beyond the stated conditional score null,*

$$\mathbb{E}_P\left[\sup_{t\ge 0} \frac{\sum_{j\in D_t} N_j}{|D_t| \vee 1}\right] \le q. \tag{8}$$

Consequently $\mathbb{E}_P[\mathrm{FDP}(D_T)] \le q$ for every almost surely finite global stopping time T, including a stopping rule that uses all market data, agent outputs, evidence, and portfolio results.

Proof. Predictability and the null condition give, on $\{N_j = 1, \tau_j < t\}$,

$$\mathbb{E}_P[E_{j,t} \mid \mathcal{F}_{t-1}] = E_{j,t-1}\{1 + \lambda_{j,t}\mathbb{E}_P[X_{j,t} \mid \mathcal{F}_{t-1}]\} \le E_{j,t-1}.$$

Thus $N_j E_{j,t}$, viewed after τ_j, is a nonnegative supermartingale starting at N_j: conditioning on $\{\tau_j = r\}$ verifies the claim in the shifted filtration $(\mathcal{F}_{\tau_j+n})_{n\ge 0}$. The promotion time S_j is global because D_t is adapted. For $\overline{E}_j = E_{j,S_j}\mathbf{1}\{S_j < \infty\}$, optional sampling at $S_j \wedge (\tau_j + m)$, followed by conditional Fatou as $m \to \infty$, yields

$$\mathbb{E}_P[N_j \overline{E}_j \mid \mathcal{F}_{\tau_j}] \le N_j. \tag{9}$$

Indeed, $N_j E_{j,S_j}\mathbf{1}\{S_j \le \tau_j + m\}$ is bounded above by the stopped variable and converges to $N_j\overline{E}_j$, which supplies the unbounded-stopping step.

For $D_t \ne \varnothing$, Eq. (7) implies pathwise

$$\mathrm{FDP}(D_t) \le q \sum_{j\in D_t} \gamma_j N_j \widetilde{E}_j \le q \sum_{j\ge 1} \gamma_j N_j \overline{E}_j.$$

The right-hand side is independent of t. Taking the supremum, applying Tonelli, conditioning on $\mathcal{F}_{\tau_j}$, and using Eq. (9) give

$$\begin{aligned}\mathbb{E}_P[\sup_t \mathrm{FDP}(D_t)] &\le q \sum_j \mathbb{E}_P[\gamma_j N_j \overline{E}_j] \\ &\le q \sum_j \mathbb{E}_P[\gamma_j N_j] \\ &\le q\mathbb{E}_P\left[\sum_j \gamma_j\right] \le q.\end{aligned}$$

The stopping-time statement follows because the FDP at T is bounded by the displayed pathwise supremum. □

The persistence mechanism has two pathwise power properties.

Proposition 3 (Containment under persistence and grid refinement). *At an inspection date t, let R_t be the one-shot floor-augmented weighted e-BH set computed from the unfrozen endpoints on the same eligible set, with the same q, weights, and floor. Then $R_t \subseteq D_t$. Moreover, if one SAFE-ALPHA run inspects on a refinement of another run's grid while all evidence paths and ledger fields remain fixed, its discovery set contains the coarser run's set at every common inspection.*

Proof. For terminal containment, take the union of the persistent run's old set and R_t. Old members pass at the old set size and terminal members pass at $|R_t|$; every member therefore passes at the no-smaller union size because the weighted boundary decreases with set size. The union is feasible for the maximal step-up rule. For grid refinement, induct on the common inspections and apply the same union argument to the refined run's old set and the coarser run's newly feasible set. □

The same argument has a portfolio interpretation. If all promoted strategies receive equal gross sleeve weights, the fraction of gross capital assigned to true nulls equals $\mathrm{FDP}(D_t)$, so its expected worst-time value is at most q. More generally, allocation shares $a_{j,t} \geq 0$ satisfy the same bound if $a_{j,t} = 0$ outside D_t, $\sum_j a_{j,t} \leq 1$, and $a_{j,t} \leq q\gamma_j \widetilde{E}_j$. A portfolio optimizer that discards or reweights certificates without preserving this inequality receives no capital-level FDR guarantee from the theorem.

The floor supplies a separate candidate-level reading. On the null branch, promotion implies $E_{j,S_j} \geq 1/q$, so conditional Markov and stopped e-validity give $P(S_j < \infty \mid \mathcal{F}_{\tau_j}) \leq q$ on $\{N_j = 1\}$. This marginal statement does not control the probability of any false promotion across the stream; that is why the weighted joint gate remains necessary.

The default telescope is robust to an unbounded stream but need not match a finite research campaign. If a cap M is declared before the first proposal, a valid finite-campaign prior is

$$\gamma_j^{(\eta)} = \frac{1-\eta}{j(j+1)} + \frac{\eta}{M}\mathbf{1}\{j \leq M\}, \qquad 0 \leq \eta < 1. \tag{10}$$

Its infinite sum is one, so Theorem 2 applies without change. For $\eta > 0$, its one-discovery boundary improves on the telescope precisely when $j(j+1) > M$ and $j \leq M$; it sacrifices the earliest ranks and makes post-cap boundaries larger by $1/(1-\eta)$. The cap is a ledger field, not the realized proposal count, and unused uniform mass is never reclaimed.

The main proof also quantifies implementation error. If stopped null evidence instead satisfies $\mathbb{E}[N_j \overline{E}_j \mid \mathcal{F}_{\tau_j}] \leq N_j c_j$, where $c_j \geq 1$ is fixed at proposal time and $\sum_j \gamma_j c_j \leq C$, the same calculation yields

$$\mathbb{E}[\sup_t \mathrm{FDP}(D_t)] \leq qC. \tag{11}$$

Thus a bounded evidence inflation has an explicit validity debt, while dividing each stream by a predictable upper bound on its cumulative inflation restores the level-q statement.

5 Empirical design

5.1 Public-data strategy stream

We use daily value-weighted returns for the 30 U.S. industry portfolios and the daily risk-free rate from the Kenneth French Data Library [5]. The report window is January 3, 2007 through May 29, 2026; five prior years initialize proposal scores. Download hashes and the complete processed panel are recorded. Because the current library reconstructs historical series after CRSP updates, this block is an auditable economic illustration, not a point-in-time claim that historical alpha was discovered.

The fixed symbolic grammar contains 89 executable candidates: one equal-weight industry-market sleeve; 48 weekly or monthly cross-industry momentum and reversal spreads; 24 monthly low- or high-volatility spreads; and 16 weekly or monthly time-series momentum or reversal overlays. Signals use 21, 63, 126, or 252 trading-day lookbacks and are lagged before execution. Long-short rules have unit gross exposure. Turnover is half the ℓ_1 change in pretrade-to-target risky weights, charged at 2, 5, or 10 basis points per unit. Risky weights drift after each return; residual capital earns the risk-free rate. Displayed sleeve allocations are restored daily from their drifted weights, and both within-sleeve and allocation turnover are charged under the declared half-ℓ_1 convention.

To isolate the statistical layer from model-specific prompting, the main experiment uses a transparent adaptive proposer. The market sleeve is a predeclared positive control assigned the first budget slot. At experiment start and on the first trading day of each subsequent quarter, the proposer selects the unproposed rule maximizing its trailing five-year annualized Sharpe minus 0.20 times its largest absolute correlation with earlier proposals. This yields 76 adaptive symbolic proposals plus the control (77 total). The replay therefore evaluates the certification layer, not the proposal quality of a particular LLM. A generative proposer can use the same interface only if every executable output is committed before its first scored payoff.

For certification, net return $R^{\mathrm{net}}_{j,t}$ becomes

$$X_{j,t} = \mathrm{clip}\left(\frac{R^{\mathrm{net}}_{j,t}}{4\max\{\widehat{\sigma}_{j,t-1}, 10^{-6}\}}, -1, 1\right),$$

where $\widehat{\sigma}_{j,t-1}$ is a lagged 63-day volatility estimate with a lagged expanding fallback. The fixed proposal-time bet is the mean of up to 1,260 available prior bounded scores divided by their second moment, clipped to $[0.01, 0.50]$; first-date histories contain 1,239 scores after scale initialization. On the frozen panel, scale multiples 2, 4, and 8 clip respectively 6.05%, 0.318%, and 0.0099% of finite scores; the corresponding fractions of proposal-time bets at the upper cap are zero, zero, and 13.0%. The primary multiple of 4 therefore keeps clipping below 0.4% without activating the upper betting cap, while the sensitivity analysis reports the two alternatives. We use Eq. (3), $q = 0.10$, and inspect every fifth trading day. At 5 and 10 basis points we rebuild candidate return, drifted-weight, and turnover paths; the proposal ledger, 2-basis-point scale reference, and bets remain fixed.

5.2 Identification of validity and power

Historical returns do not reveal which nulls are true. We therefore separate calibration from economics.

Cross-sectional sign null. For each of 500 replications, an independent Rademacher sign ε_t multiplies the entire 89-dimensional realized score vector x_t on each date. This leaves $x_t x_t^{\mathsf{T}}$, and hence all same-date cross-strategy dependence and heteroskedastic magnitudes, unchanged. Because ε_t is independent of the signed past, $\mathbb{E}[\varepsilon_t x_{j,t} \mid \mathcal{F}_{t-1}] = 0$, even when the candidate index was selected adaptively from that past. The experiment does not preserve serial sign predictability and is not presented as a complete market simulator. The proposer reselects adaptively every 63 trading days after assigning the signed market control to the first slot, until the ledger contains 77 proposals. We compare SAFE-ALPHA with (i) an illegal same-bar bettor that wagers only after seeing that the current score is nonnegative, (ii) one-sided 5% t-tests inspected every week after 63 observations, and (iii) one unadjusted one-sided 5% test after a 252-day hold-out for each horizon-eligible proposal (74 of 77). The last comparator diagnoses ignored cross-proposal multiplicity; it is not a multiplicity-adjusted fixed split.

The wild-sign construction preserves realized same-date dependence and volatility magnitudes but removes serial sign structure. We therefore add a fixed synthetic null with standardized Student t_3 innovations, a predictable common GARCH-style scale recursion, cross-candidate correlation 0.5, and clipping to the same score range. In both null designs every procedure sees the identical candidate paths. A nine-point grid from $q = 0.01$ through 0.20 describes sensitivity around $q = 0.10$ on the same paths.

Paired planted alternatives. Each replication has 80 bounded Gaussian-score candidates, of which 16 have positive drift, 504 burn-in days, and a common 1,260-day proposal/evaluation period. The first proposal has the largest burn-in sample mean; every later proposal is the remaining candidate with the largest trailing annualized Sharpe, with no diversity penalty. Up to 50 candidates are selected at 15-day intervals, so strict post-proposal monitoring ranges from 1,259 days for the earliest proposal to 524 for the latest. The reported planted Sharpe is the nominal pre-clipping mean-to-volatility ratio. It ranges from 0.50 to 3.00 in increments of 0.25, at equicorrelations zero and 0.5, with 250 replications in every cell. SAFE-ALPHA and the matched terminal weighted e-BH comparator receive the same scores, proposal order, proposal-time bets, spending weights, and raw e-process paths; only the decision schedule differs. The terminal rule reads the untouched endpoints once, whereas SAFE-ALPHA inspects every five days and freezes only promoted copies. Ordinary terminal BH, uncorrected 5% tests, and the five largest post-proposal sample means remain diagnostic comparators. We report realized FDP, end-to-end power over all 16 planted alternatives, proposal recall, conditional gate power, and delay from proposal to decision.

Prospective power extension. No market outcome entered power-design selection. On two recorded 300-path development banks, we chose Eq. (10) with $M = 50$, $\eta = .75$, daily inspection, and the equal arithmetic mixture of three e-processes whose proposal-time bets are $\min\{.5, m\widehat{\lambda}_j\}$ for $m \in \{1, 2, 4\}$. Such fixed mixtures are established betting constructions, not a novelty claim [10, 17]. We froze this design and then generated six new 250-path cells at Sharpes 1, 1.5, and 2 and correlations zero and .5. The original public replay remains on the telescope, the fixed plug-in bet, and five-day inspection. A valid conservative comparator promotes proposal j only when $E_{j,t} \geq (q\gamma_j)^{-1}$; conditional Ville and $\sum_j \gamma_j \leq 1$ give FWER at most q. Every comparator receives the same score paths, proposal order, weights, betting components, and inspection grid permitted by the comparison.

Economic and selection diagnostics. We report an ungated top-five policy reselected quarterly from the trailing five years; a fixed split trained from 2002 through 2010, validated from 2011 through 2014, and tested from 2015 through 2026; and a deliberately optimistic full-sample top five. For the certified sleeve we estimate post-certification CAPM and Fama-French five-factor plus momentum intercepts by OLS with heteroskedasticity and autocorrelation consistent standard errors using five lags. A second gate predictably removes those six factor returns with rolling betas estimated only from the previous 1,260 dates and at least 252 observations. Across the 74 symbolic proposals with a complete one-year future window, we compare the five-year selection Sharpe with the next-year Sharpe and bootstrap the median change. Finally, we compute White's Reality Check, Hansen's SPA with consistent recentering, 1,999 stationary-bootstrap draws, and mean block length 20, the Deflated Sharpe Ratio, and ten-block CSCV/PBO. The Reality Check retains its circular-block convention, so its p-values are not numerically interchangeable with SPA. These fixed-search diagnostics test economic interpretations that the sequential score certificate itself does not imply.

6 Results

6.1 Calibration, persistence, and power

The original procedure makes at least one false promotion in 20 of 500 same-date sign-null runs, probability 0.040 with 95% Wilson interval $[0.026, 0.061]$. Across nine levels, the curve rises from 0.006 at $q = .01$ to 0.072 at $q = .20$ and remains below the diagonal. Under the final campaign and betting design, the corresponding $q = .10$ probability is 0.042, interval $[0.028, 0.063]$, versus 0.040 for proposal-wise FWER. The heavy-tailed volatility null gives 0.034 for the original gate and 0.020 for the final design. By contrast, same-bar

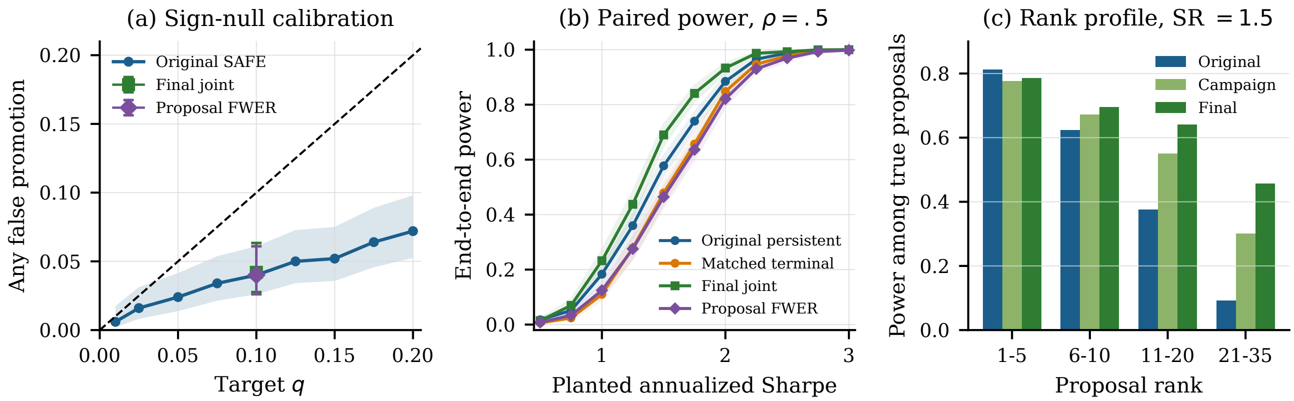

Figure 1: Calibration and paired power. Panel (a) uses 500 sign-null paths; the curve is the original protocol and the two $q = .10$ points use the final design. Panel (b) uses 250 paired paths per effect at $\rho = .5$; shaded regions are seed-level 95% intervals. Panel (c) uses the newly seeded central holdout and counts only true proposals within each rank bin.

betting and repeatedly inspected 5% tests promote in every sign-null run; separate one-year tests across 74 proposals promote in 94.4%. These gray-box ablations identify timing and multiplicity failures, whereas the terminal e-BH and proposal-wise FWER procedures are valid comparators.

At annualized Sharpe 1.5 and $\rho = .5$, the original persistent rule has power 0.578, versus 0.479 for matched terminal weighted e-BH. The paired gain is 0.099, with interval $[0.084, 0.113]$, and persistence never loses on any of the 250 matched paths, as Proposition 3 predicts. Mean decision delay among detected alternatives is 684 trading days for persistence and 1,183 for terminal analysis; the detected subsets differ, so this delay contrast is descriptive rather than a paired causal estimate.

The newly seeded holdout confirms the prospective power extension. At the same central effect, the final design reaches 0.693, compared with 0.645 for campaign weights and daily inspection alone and 0.572 for the original schedule. Its paired gains are 0.048, interval $[0.034, 0.062]$, over the campaign-only design and 0.121, interval $[0.104, 0.137]$, over the original. Against proposal-wise e-alpha-spending on a dense paired grid, joint power is 0.690 versus 0.465, a gain of 0.226 with interval $[0.208, 0.243]$; the joint gate wins on 225 paths and ties on 25. The gain is rank-specific. Relative to the original, final power changes from 0.812 to 0.785 at ranks 1–5, but rises from 0.623 to 0.695 at ranks 6–10, from 0.375 to 0.641 at ranks 11–20, and from 0.091 to 0.457 at ranks 21–35. This is the intended finite-campaign tradeoff, not uniform dominance.

6.2 Locked public replay and selection decay

The rank-one market sleeve is a prespecified pipeline control rather than an adaptively discovered alpha. It crossed its boundary on January 19, 2018 at frozen evidence 20.087, showing that the proposal, evidence, promotion, and next-bar deployment path can operate on financial data. The substantive discovery result is negative: none of 76 adaptively proposed symbolic rules was certified at $q = 0.10$, and predictable six-factor residualization produced no certificate at all. We therefore read the replay as evidence about selection control in adaptive strategy search, not as evidence that SAFE-ALPHA discovers factor-adjusted alpha. The market control still crossed at 5 and 10 basis points, with the 10-basis-point crossing delayed by one week.

The ledger entry makes the joint rule concrete. At the January 19 inspection, the market control had $\gamma_1 = 1/2$, so a one-member discovery set required

$$E_{1,t} \geq \max\{1/q, (q\gamma_1)^{-1}\} = \max\{10, 20\} = 20.$$

Its current value was 20.087, while no other eligible candidate supported a larger feasible set. The gate froze that value, recorded the code and cost specification used to obtain it, and began portfolio deployment on the next tradable date rather than earning the triggering return twice. Figure 2 records the corresponding weighted paths.

Deployment begins one trading day after promotion. From January 22, 2018 through May 29, 2026, the market control earns 9.96% annualized net excess return at 19.77% volatility, Sharpe 0.504, and maximum drawdown −38.70%. These are properties of a broad market sleeve, not evidence of mined alpha; the large drawdown also shows why a score certificate is not a risk guarantee.

The factor evidence in Table 2 clarifies the certificate's interpretation. Over 2,100 post-certification dates, the market sleeve has a CAPM beta of 0.941; the six-factor regression explains 95.9% of its variation and estimates a market beta of 0.948. Both intercepts are negative and statistically insignificant. When candidate returns are predictably residualized with betas fitted only through the preceding date, the gate

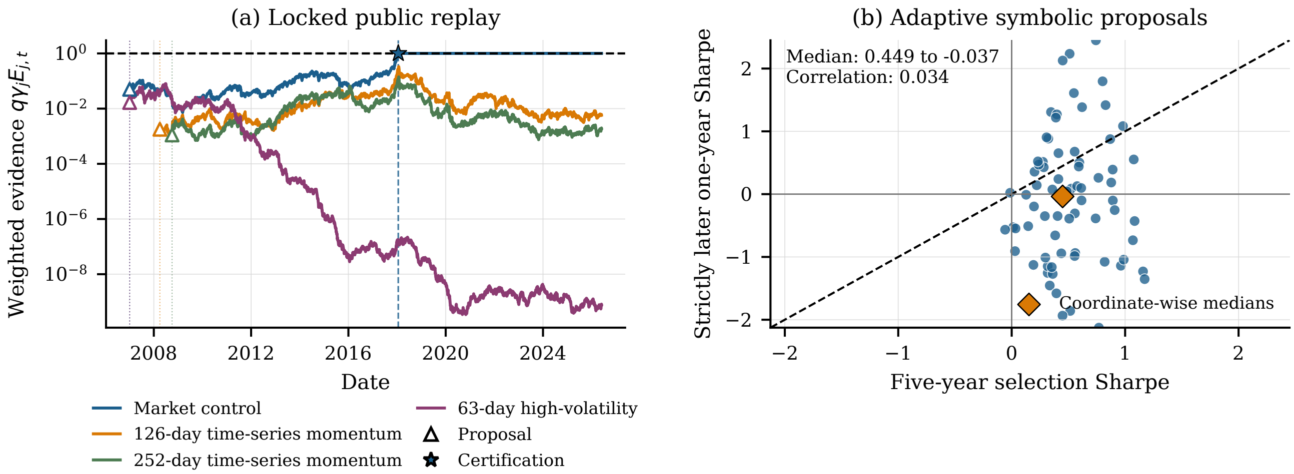

Figure 2: Locked public replay and selection decay. Panel (a) initializes each path at unit evidence on its proposal date and updates it strictly later; the market path is frozen at its crossing. The three symbolic paths are selected for display by their recorded maxima, which never enter the gate. Panel (b) uses the 74 symbolic proposals with a complete 252-day future window; the diamond marks the coordinate-wise medians.

Table 2: Discovery and fixed-family diagnostics. "All" includes the prespecified market control. Reality Check and SPA test different objects from the sequential conditional-score null.

Diagnostic	Adaptive	All
Primary certificates	0 of 76	1 of 77
Factor-residual certificates	0 of 76	0 of 77
Median selected / future SR	.449 / -.037	—
Reality Check p	.190	.022
SPA p	.146	.157

certifies no strategy. The sole primary certificate is therefore a broad market positive control under the committed bounded score, not evidence of factor-adjusted alpha.

The proposal ledger displays the selection effect that a terminal winner table would conceal. Among 74 symbolic proposals with a complete 252-day future window, median annualized Sharpe falls by 0.512, 70.3% perform worse after proposal, only 48.6% remain positive, and the correlation between selection and future Sharpe is 0.034. An independent proposal-resampling 95% percentile interval is $[0.398, 0.754]$, but it is descriptive because the proposals overlap. White's Reality Check rejects on the full 89-rule panel, but not after removing the market control or restricting attention to adaptive proposals; consistently recentered SPA does not reject in either the full or adaptive family. Although these procedures test different nulls, they point in the same descriptive direction once the prespecified market sleeve is distinguished from the mined symbolic rules.

One-at-a-time design changes reveal where the negative result is stable and where the estimand changes. At $q = 0.05$ no candidate is certified; at $q = 0.20$ the market control and one monthly time-series momentum rule cross. Daily inspection preserves the primary market certificate, inspection every 21 days misses its transient valid crossing, and moving the market control from budget rank one to ranks 5, 10, or 20 produces no certification. Scaling returns by two lagged volatilities yields three promotions, while scaling by eight yields none. These are not interchangeable tuning choices after the fact: they define different ledger-fixed scores, budgets, and inspection opportunities, which is why the ledger records them explicitly.

7 Audit and replication record

The released strategy catalog and proposal, evidence, and certification ledgers identify every empirical decision, while tests enforce strict post-proposal updating and next-bar deployment. SHA-256 digests cover inputs, code, and reported outputs. Replication-level records retain the seed map for the 5,500 paired evaluation paths together with null, sensitivity, and portfolio accounting results.

8 Limitations and scope

The theorem applies to Eq. (1), not to a weaker unconditional alpha null. Our bounded transformation changes the estimand, and a production system seeking a raw-return claim should instead use an e-process justified by explicit tail or conditional-moment assumptions. FDR controls membership of the certified set; it does not guarantee profitability, diversification, market capacity, CVaR, or low drawdown. A downstream optimizer must obey the allocation constraint above if capital-weighted false exposure is the target.

The deterministic proposer is an identification choice: the study isolates the gate rather than benchmarking language models, so it does not establish how proposal quality changes under a particular LLM. The public industry series are nontradable research portfolios and not archival point-in-time vintages; costs omit impact, borrowing, and capacity. Planted alternatives impose persistent constant drift in a stationary Gaussian-score design, not alpha decay, regime change, or crowding. Finally, every summable rank prior trades power across proposal positions. The finite-campaign schedule improves middle and late ranks within its declared cap but weakens the earliest ranks and the post-cap stream. Weak persistent effects can still require long monitoring histories.

9 Conclusion

SAFE-ALPHA reserves promotion for a statistical gate that reads only post-proposal payoffs and freezes evidence at a joint crossing. Under the stated conditional null, the persistent set controls worst-time expected FDP despite dependence and adaptive stopping, while a declared campaign prior, daily inspection, and a fixed betting mixture raise central holdout power by 12.1 percentage points relative to the original schedule. The public replay reaches a deliberately restrained conclusion: the prespecified market control crosses, none of 76 adaptive symbolic rules does, and factor residualization removes the control certificate. The practical value is therefore not a promise of profitable alpha, but a defensible separation between attractive research output and evidence strong enough to justify deployment.

A Formal Protocol and Statistical Estimand

This appendix gives the complete probability model and proofs underlying the SAFE-ALPHA certificate. Time is discrete. Let $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\geq 0}, P)$ be a filtered probability space, where $\mathcal{F}_t$ is the *global* information available to the research system by time t. It contains all market observations, data releases, agent inputs and outputs, candidate specifications, execution records, evidence streams, and gate decisions observed by then. For a stopping time τ, write

$$\mathcal{F}_\tau = \{B \in \mathcal{F} : B \cap \{\tau \leq t\} \in \mathcal{F}_t \text{ for every } t\}.$$

Candidate j arrives at an $(\mathcal{F}_t)$-stopping time τ_j. Its executable specification A_j is $\mathcal{F}_{\tau_j}$-measurable. The specification includes the signal formula, universe, data vintage, execution lag, missing-data convention, position and turnover definitions, cost rule, score transformation, and evidence rule. We index only candidates that enter the protocol, so $\tau_j < \infty$ almost surely. There may be infinitely many candidates over the life of the system, but only finitely many may have arrived by any finite calendar time. A substantive revision of A_j receives a new index, a new proposal time, and unit initial evidence.

For $t > \tau_j$, let $X_{j,t}$ be the score produced by the committed strategy. It is $\mathcal{F}_t$-measurable, and for the linear betting construction below we assume

$$-1 \leq X_{j,t} \leq 1 \quad \text{almost surely.} \tag{12}$$

For fixed P, let $N_j \in \{0,1\}$ indicate that the randomly selected candidate is a true null. We require N_j to be $\mathcal{F}_{\tau_j}$-measurable and take the following selective conditional-drift restriction as the primitive null condition:

$$N_j \mathbf{1}\{\tau_j < t\} \mathbb{E}_P[X_{j,t} \mid \mathcal{F}_{t-1}] \leq 0 \quad \text{almost surely for every } t. \tag{13}$$

Thus the identity of a null may be selected from the global past. What is excluded is selection with information from the outcome that will test it. One formal construction starts with a countable or standard-Borel strategy grammar $\mathcal{S}$. Each $a \in \mathcal{S}$ has a null model $\mathcal{P}_a$, A_j is selected at τ_j, and $N_j(P) = \mathbf{1}\{P \in \mathcal{P}_{A_j}\}$. In that construction, membership in $\mathcal{P}_a$ must imply Eq. (13) uniformly over every admissible adaptive history that can lead to $A_j = a$. Marginal mean zero for each strategy when prespecified does not suffice.

A.1 What is certified

The theorem certifies the estimand represented by $X_{j,t}$, not an unstated raw-return estimand. To make this distinction explicit, suppose the committed position earns gross excess return $G_{j,t}$, its turnover is

$$U_{j,t} = \frac{1}{2}\|w_{j,t} - w_{j,t^-}\|_1,$$

and the ledger charges $c_{j,t}$ per unit of turnover. The tested net return is

$$R^{\text{net}}_{j,t} = G_{j,t} - c_{j,t}U_{j,t}. \tag{14}$$

The position, execution rule, and accounting convention must be fixed before the corresponding return is realized. A conservative cost schedule may be fixed at proposal time; it may also be updated predictably, with $c_{j,t} \in \mathcal{F}_{t-1}$. Actual fills and realized fees can instead enter Eq. (14) through a precommitted accounting rule. Choosing the cost charge after inspecting $G_{j,t}$ is not admissible.

In the empirical protocol, a predictable scale $b_{j,t} > 0$ is formed from lagged volatility and

$$X_{j,t} = \operatorname{clip}\left(\frac{R^{\text{net}}_{j,t}}{b_{j,t}}, -1, 1\right), \qquad b_{j,t} = 4\max\{\widehat{\sigma}_{j,t-1}, 10^{-6}\}, \tag{15}$$

where $\widehat{\sigma}_{j,t-1}$ is estimated from 63 lagged observations. A lagged expanding estimate is used during initialization. The null concerns the conditional mean of this bounded net score. In general,

$$\mathbb{E}[R^{\text{net}}_{j,t} \mid \mathcal{F}_{t-1}] \leq 0 \not\Longrightarrow \mathbb{E}[\operatorname{clip}(R^{\text{net}}_{j,t}/b_{j,t}, -1, 1) \mid \mathcal{F}_{t-1}] \leq 0.$$

Consequently, the certificate must not be reported as a test of raw mean return. A raw-mean interpretation requires an actually bounded payoff implemented through position limits, a known one-sided payoff bound, or a different e-process justified by a stated conditional moment-generating function or variance assumption. The multiple-testing theorem below is agnostic about that choice: it uses only the selective e-validity property proved next.

B Global E-Validity After Adaptive Proposal

Let $\lambda_{j,t} \in [0,1]$ be $\mathcal{F}_{t-1}$-measurable. The bet may use every element of the global past, including other candidates' histories, but not the current payoff. Define

$$E_{j,t} = \begin{cases} 1, & t \le \tau_j, \\ \prod_{s=\tau_j+1}^{t} (1 + \lambda_{j,s} X_{j,s}), & t > \tau_j. \end{cases} \tag{16}$$

Every factor is nonnegative by Eq. (12).

Lemma 4 (Global e-validity). *Suppose Eq.* (13) *holds. Let $S \ge \tau_j$ be any global stopping time, possibly unbounded or infinite, and define its killed stopping value by*

$$\overline{E}_{j,S} = E_{j,S}\mathbf{1}\{S < \infty\},$$

with the product in Eq. (16) *evaluated only on $\{S < \infty\}$. Then*

$$\mathbb{E}_P[N_j \overline{E}_{j,S} \mid \mathcal{F}_{\tau_j}] \le N_j \quad \textit{almost surely.} \tag{17}$$

In particular, $\mathbb{E}_P[N_j \overline{E}_{j,S}] \le \mathbb{E}_P[N_j]$.

Proof. First consider one post-proposal update. On the event $N_j = 1$ and $\tau_j < t$, predictability and Eq. (13) give

$$\begin{aligned} \mathbb{E}_P[E_{j,t} \mid \mathcal{F}_{t-1}] &= E_{j,t-1}\mathbb{E}_P[1 + \lambda_{j,t} X_{j,t} \mid \mathcal{F}_{t-1}] \\ &= E_{j,t-1}\left\{1 + \lambda_{j,t}\mathbb{E}_P[X_{j,t} \mid \mathcal{F}_{t-1}]\right\} \le E_{j,t-1}. \end{aligned} \tag{18}$$

When $N_j = 0$, multiplication by N_j makes both sides zero.

The random starting time requires a short argument. Set $\mathcal{G}_{j,n} = \mathcal{F}_{\tau_j+n}$ for $n \ge 0$. Because $\tau_j + n$ is a stopping time, this is a filtration, and $N_j E_{j,\tau_j+n}$ is $\mathcal{G}_{j,n}$-measurable. If $B \in \mathcal{G}_{j,n-1}$, then $B \cap \{\tau_j = r\} \in \mathcal{F}_{r+n-1}$. Partitioning over the possible values of τ_j and using Eq. (18) yields

$$\begin{aligned} \mathbb{E}_P[\mathbf{1}_B N_j E_{j,\tau_j+n}] &= \sum_{r\ge 0} \mathbb{E}_P[\mathbf{1}_{B\cap\{\tau_j = r\}} N_j E_{j,r+n}] \\ &\le \sum_{r\ge 0} \mathbb{E}_P[\mathbf{1}_{B\cap\{\tau_j = r\}} N_j E_{j,r+n-1}] \\ &= \mathbb{E}_P[\mathbf{1}_B N_j E_{j,\tau_j+n-1}]. \end{aligned}$$

It follows that

$$(N_j E_{j,\tau_j+n})_{n\ge 0}$$

is a nonnegative $(\mathcal{G}_{j,n})$-supermartingale. Its initial value is N_j.

The shifted time $R = S - \tau_j$ is a nonnegative $(\mathcal{G}_{j,n})$-stopping time: for every n, $\{R \le n\} = \{S \le \tau_j + n\} \in \mathcal{G}_{j,n}$. Conditional optional sampling at $R \wedge m$ therefore gives

$$\mathbb{E}_P[N_j E_{j,S\wedge(\tau_j+m)} \mid \mathcal{F}_{\tau_j}] \le N_j. \tag{19}$$

Now multiply the stopped variable by $\mathbf{1}\{S \le \tau_j + m\}$. This can only decrease it, and the resulting nonnegative variables converge pointwise to $N_j\overline{E}_{j,S}$: if $S < \infty$, equality with $N_j E_{j,S}$ holds for all sufficiently large m; if $S = \infty$, they are identically zero. Conditional Fatou's lemma applied to Eq. (19) proves Eq. (17). Taking expectations proves the final statement. □

The product in Eq. (16) is convenient rather than essential. For example, if under the null

$$\mathbb{E}_P[\exp\{\eta_{j,t} X_{j,t} - \eta_{j,t}^2 v_{j,t}/2\} \mid \mathcal{F}_{t-1}] \le 1$$

for predictable $\eta_{j,t}$ and $v_{j,t}$, the corresponding exponential product is a global test supermartingale. A convex mixture of global e-processes is again globally valid. Such constructions can replace clipping when their conditional assumptions are economically defensible.

C The Irreversible Weighted Gate

Fix $q \in (0,1)$. Candidate j receives a nonnegative weight $\gamma_j \in \mathcal{F}_{\tau_j}$ such that

$$\sum_{j=1}^{\infty} \gamma_j \le 1 \quad \text{almost surely.} \tag{20}$$

The deterministic choice $\gamma_j = 1/\{j(j+1)\}$ telescopes to one and accommodates an unbounded proposal stream. A proposal-time, data-dependent allocation is also permitted, provided Eq. (20) holds pathwise. The convention $1/\gamma_j = \infty$ makes a zero-weight proposal ineligible for promotion.

At inspection t, let $\mathcal{A}_t$ be the finite, $\mathcal{F}_t$-measurable set of eligible candidates satisfying

$$D_{t-1} \subseteq \mathcal{A}_t \subseteq \{j : \tau_j < t\}. \tag{21}$$

Eligibility may include a prespecified post-proposal seasoning period, but an unproposed index can never enter the gate. Initialize $D_{-1} = \varnothing$, and suppose inductively that D_{t-1} is finite and that each $j \in D_{t-1}$ has a frozen value

$$\widetilde{E}_j = E_{j,S_j}, \qquad S_j = \inf\{s \ge \tau_j : j \in D_s\}.$$

Form

$$Z_{j,t} = \begin{cases} \widetilde{E}_j, & j \in D_{t-1}, \\ E_{j,t}, & j \in \mathcal{A}_t \setminus D_{t-1}. \end{cases} \tag{22}$$

Set the mandatory individual evidence floor to $h_q = q^{-1}$. For $k \ge 1$, let

$$C_t(k) = \sum_{j\in\mathcal{A}_t} \mathbf{1}\left\{ Z_{j,t} \ge \max\left(h_q, \frac{1}{q\gamma_j k}\right)\right\}. \tag{23}$$

and define

$$k_t^\star = \max\left(\{0\} \cup \{k \in \{1,\ldots,|\mathcal{A}_t|\} : C_t(k) \ge k\}\right), \tag{24}$$

$$D_t = \begin{cases} \left\{ j \in \mathcal{A}_t : Z_{j,t} \ge \max\left(h_q, (q\gamma_j k_t^\star)^{-1}\right)\right\}, & k_t^\star > 0, \\ \varnothing, & k_t^\star = 0. \end{cases} \tag{25}$$

Every new member $j \in D_t \setminus D_{t-1}$ receives $S_j = t$ and $\widetilde{E}_j = E_{j,t}$.

Proposition 5 (Well-definedness, irreversibility, and self-consistency). *The recursion in Eqs.* (22) *and* (25) *is adapted and finite. For every t,*

$$D_{t-1} \subseteq D_t, \qquad |D_t| = k_t^\star, \tag{26}$$

and

$$j \in D_t \implies \widetilde{E}_j \ge h_q, \qquad \widetilde{E}_j \ge \frac{1}{q\gamma_j |D_t|}. \tag{27}$$

Consequently, each S_j is a global stopping time.

Proof. All operations at inspection t involve a finite collection of $\mathcal{F}_t$-measurable variables, so $k_t^\star$ and D_t are $\mathcal{F}_t$-measurable. It remains to establish the invariants. Proceed by induction. They are immediate at initialization. Write $d = |D_{t-1}|$. If $d > 0$, the induction hypothesis gives, for every old member,

$$\widetilde{E}_j \geq \max\left(h_q, \frac{1}{q\gamma_j d}\right).$$

Hence $C_t(d) \geq d$, so d is feasible in Eq. (24). If $d = 0$, feasibility of $k = 0$ is built into the definition. In either case $k_t^\star \geq d$. The map $k \mapsto \max\{h_q, (q\gamma_j k)^{-1}\}$ is nonincreasing, so every old member exceeds its threshold at $k_t^\star$. Thus $D_{t-1} \subseteq D_t$.

If $k_t^\star = 0$, Eq. (25) is empty and its cardinality equals $k_t^\star$. Now suppose $1 \leq k_t^\star < |\mathcal{A}_t|$ and $C_t(k_t^\star) > k_t^\star$. The threshold weakly decreases when k increases, so

$$C_t(k_t^\star + 1) \geq C_t(k_t^\star) \geq k_t^\star + 1.$$

This contradicts maximality. If $k_t^\star = |\mathcal{A}_t|$, equality between the count and $k_t^\star$ is automatic. Therefore the set in Eq. (25) has exactly $k_t^\star$ members. Its old members retain their frozen values, and every new member is frozen at the value used for admission. This proves Eq. (27).

Finally, nestedness and Eq. (21) give $\{S_j \leq t\} = \{j \in D_t\} \in \mathcal{F}_t$. Hence S_j is a global stopping time. □

The floor is not required for joint FDR control, but it rules out a weak-evidence passenger after other discoveries have lowered the step-up boundary. On the null branch, stopped e-validity and conditional Markov give $P(S_j < \infty \mid \mathcal{F}_{\tau_j}) \leq q$. This is a marginal statement for candidate j, not family-wise error control for the proposal stream.

D Simultaneous False-Discovery Control

The next result permits arbitrary dependence, distinct monitoring horizons, adaptive proposal times, and infinitely many eventual proposals. Its multiple-testing step is the weighted self-consistency principle behind e-BH and online e-BH [4, 16, 20]; the proof must also handle random strategy identities and global stopping of evolving evidence.

Theorem 6 (Persistent SAFE-ALPHA certificate). *For each j, suppose:*

1. *τ_j is a global stopping time and the candidate identity and null indicator are $\mathcal{F}_{\tau_j}$-measurable;*
2. *$E_{j,t}$ has the selective global e-validity property (17) at every global stopping time $S \geq \tau_j$;*
3. *$\gamma_j \geq 0$ is $\mathcal{F}_{\tau_j}$-measurable and the pathwise budget (20) holds; and*
4. *D_t is a finite, adapted, nested discovery process satisfying Eq. (27).*

Define

$$\mathrm{FDP}(D_t) = \frac{\sum_{j \in D_t} N_j}{|D_t| \vee 1}.$$

Then, under arbitrary serial and cross-candidate dependence,

$$\mathbb{E}_P\left[\sup_{t\geq 0} \mathrm{FDP}(D_t)\right] \leq q. \tag{28}$$

Consequently,

$$\mathbb{E}_P[\mathrm{FDP}(D_T)] \leq q \tag{29}$$

for every almost surely finite global stopping time T. The rule defining T may use the complete research history, including market returns, agent outputs, all evidence streams, portfolio performance, and prior gate decisions.

Proof. Define the killed promotion value

$$\overline{E}_j = E_{j,S_j} \mathbf{1}\{S_j < \infty\}.$$

Proposition 5 makes S_j a global stopping time, and selective e-validity gives

$$\mathbb{E}_P[N_j \overline{E}_j \mid \mathcal{F}_{\tau_j}] \leq N_j. \tag{30}$$

Fix a sample path and a time t. If $D_t = \varnothing$, its FDP is zero. Otherwise, Eq. (27) implies, for every j,

$$\frac{\mathbf{1}\{j \in D_t\}}{|D_t|} \leq q\gamma_j \widetilde{E}_j \mathbf{1}\{j \in D_t\}.$$

Multiplication by N_j, followed by summation, gives the pathwise bound

$$\mathrm{FDP}(D_t) \leq q \sum_{j \in D_t} \gamma_j N_j \widetilde{E}_j \leq q \sum_{j=1}^{\infty} \gamma_j N_j \overline{E}_j. \tag{31}$$

The last expression is independent of t. Therefore

$$\sup_{t\geq 0} \mathrm{FDP}(D_t) \leq q \sum_{j=1}^{\infty} \gamma_j N_j \overline{E}_j.$$

All summands are nonnegative. Tonelli's theorem, proposal-time measurability of γ_j, and Eq. (30) now yield

$$\begin{aligned}
\mathbb{E}_P[\sup_t \mathrm{FDP}(D_t)] &\leq q \sum_{j=1}^{\infty} \mathbb{E}_P[\gamma_j N_j \overline{E}_j] \\
&= q \sum_{j=1}^{\infty} \mathbb{E}_P\left[\gamma_j \, \mathbb{E}_P[N_j \overline{E}_j \mid \mathcal{F}_{\tau_j}]\right] \\
&\leq q \sum_{j=1}^{\infty} \mathbb{E}_P[\gamma_j N_j] \\
&\leq q \, \mathbb{E}_P\left[\sum_{j=1}^{\infty} \gamma_j\right] \leq q.
\end{aligned}$$

This argument is unaffected by infinitely many eventual candidates: finiteness at each inspection makes the gate well defined, and Tonelli justifies the countable sum. Finally, the pathwise inequality

$$\mathrm{FDP}(D_T) \leq \sup_t \mathrm{FDP}(D_t)$$

proves Eq. (29). □

The supremum appears *inside* the expectation in Eq. (28); this is stronger than fixed-time FDR control and immediately implies control under optional stopping. The argument depends on freezing a single e-valid value at each global promotion time. It does not treat a candidate's raw historical maximum as an e-value.

PROPOSITION 7 (CONTAINMENT OF MATCHED TERMINAL E-BH). *At an inspection date t, let R_t be the one-shot floor-augmented weighted e-BH set obtained from the unfrozen endpoints $E_{j,t}$ on $\mathcal{A}_t$, using the same q, γ, and h_q as the persistent gate. Then $R_t \subseteq D_t$ pathwise.*

PROOF. Write $k = |R_t|$ and $d = |D_{t-1}|$. Every terminal member passes the floor and its size-k boundary. By Eq. (27), every old member passes the floor and its size-d boundary; every candidate outside D_{t-1} uses the same raw endpoint in both procedures. If $k \leq d$, all terminal members also pass at size d, and $k_t^\star \geq d$. If $k > d$, the union of the old and terminal sets makes size k feasible, so $k_t^\star \geq k$. In either case, the persistent threshold at $k_t^\star$ is no larger than the boundary already passed by each member of R_t, proving the inclusion. □

E The Global-Filtration Condition and Its Boundaries

E.1 A causal sufficient condition

Equation (13) is the exact condition used in the proof. The following representation gives it an operational interpretation. Let $C_t \in \mathcal{F}_{t-1}$ be a recorded market state and let Y_t be the vector of contemporaneous innovations. Suppose a transition kernel K_t satisfies

$$\mathcal{L}(Y_t \mid \mathcal{F}_{t-1}) = K_t(\cdot \mid C_t). \tag{32}$$

The committed strategy produces

$$X_{j,t} = g(A_j, C_t, Y_t).$$

Because $A_j \in \mathcal{F}_{t-1}$ whenever $t > \tau_j$, the null condition

$$N_j \int g(A_j, C_t, y)\, K_t(dy \mid C_t) \leq 0 \quad \text{on } \{\tau_j < t\} \tag{33}$$

implies Eq. (13) by conditional integration. The coordinates of Y_t, and hence all strategy returns on the same date, may be arbitrarily dependent. The kernel conditions in Eqs. (32) and (33) do not assert a Markov market. They isolate the relevant causal restriction: no unrecorded future innovation may reach the proposer, another evidence stream, or the gate before the tested payoff is realized. This is the global-filtration condition emphasized for stopped e-BH by Wang et al. [15].

The condition translates into auditable implementation rules:

(1) a signal using the close at t trades no earlier than the first implementable price after that close;
(2) proposal timestamps, data vintages, code hashes, random seeds, cost inputs, and execution lags are committed before evaluation;
(3) pre-proposal backtest returns never enter $E_{j,t}$;
(4) a formula, universe, lag, or regime revision starts a new candidate;
(5) all evidence streams are aligned to the same physical market clock, rather than by observation number; and
(6) only the betting rule, not the tested payoff formula, may adapt predictably within an existing evidence process.

E.2 Same-bar leakage

Let X be a fair Rademacher payoff. It satisfies the null $\mathbb{E}[X] = 0$. If a bettor illegally observes the same-bar payoff and then sets

$$\lambda = \tfrac{1}{2}\mathbf{1}\{X = 1\},$$

the purported one-step evidence multiplier is $3/2$ when $X = 1$ and 1 when $X = -1$. Hence

$$\mathbb{E}[1 + \lambda X] = \tfrac{1}{2}(3/2) + \tfrac{1}{2}(1) = \tfrac{5}{4} > 1.$$

The loss of validity is entirely due to nonpredictability. The example also shows why a signal computed with a closing price cannot be credited with a trade at that same closing price.

E.3 Local validity does not imply global validity

Let $Y_1^1, Y_2^1, \dots$ be independent Rademacher variables, and define a second stream by $Y_n^2 = Y_{n+1}^1$. In the filtration generated only by the first stream,

$$M_n^1 = \prod_{s=1}^{n} (1 + Y_s^1/2)$$

is a nonnegative martingale. The global filtration containing both streams reveals Y_{n+1}^1 at time n. In that filtration,

$$S = 1 + \mathbf{1}\{Y_1^2 = 1\} = 1 + \mathbf{1}\{Y_2^1 = 1\}$$

is a stopping time. Nevertheless,

$$\begin{aligned}\mathbb{E}[M_S^1] &= \tfrac{1}{2}\mathbb{E}[M_2^1 \mid Y_2^1 = 1] + \tfrac{1}{2}\mathbb{E}[M_1^1 \mid Y_2^1 = -1] \\ &= \tfrac{1}{2}(3/2) + \tfrac{1}{2}(1) = \tfrac{5}{4}.\end{aligned}$$

Thus local optional-stopping validity is insufficient for a gate that observes all streams. This counterexample is the basic leakage mechanism identified by Wang et al. [15]. SAFE-ALPHA requires every candidate's e-process to be valid in the filtration in which the gate stops it.

E.4 A random terminal family defeats ordinary e-BH

Summable proposal weights are not a cosmetic replacement for the terminal number of candidates. Let all hypotheses be null and, independently for $j \geq 1$, set

$$E_j = \begin{cases} j/q, & \text{with probability } q/j, \\ 0, & \text{otherwise.} \end{cases}$$

Each E_j is an e-value because $\mathbb{E}[E_j] = 1$. Stop proposing at

$$J = \inf\{j : E_j = j/q\}.$$

Since $\sum_j q/j = \infty$, independence implies $J < \infty$ almost surely. Ordinary unweighted e-BH applied to the random family $E_1, \dots, E_J$ rejects H_J: for one rejection its threshold is J/q, exactly E_J. It therefore has FDR one. A deterministic maximum M is safe with $\gamma_j = 1/M$ for all M slots, including unused slots. An unbounded adaptive stream requires a summable budget such as Eq. (20).

E.5 A raw running maximum is not an e-value

Consider the one-step martingale $E_0 = 1$ and

$$E_1 = \begin{cases} 2, & \text{with probability } 1/2, \\ 0, & \text{with probability } 1/2, \end{cases}$$

held constant thereafter. Although $\mathbb{E}[E_1] = 1$,

$$\mathbb{E}\left[\max_{s\le 1} E_s\right] = \tfrac{1}{2}(2) + \tfrac{1}{2}(1) = \tfrac{3}{2}.$$

The running maximum is therefore not an e-value. Under dependence, submitting raw e-process maxima to e-BH can inflate FDR; explicit multiple-testing counterexamples are given by Tavyrikov et al. [13]. SAFE-ALPHA does something different. It freezes $E_{j,t}$ only when the *current* value passes the joint self-consistent gate. That first-promotion time is global, and Lemma 4 validates the single frozen value. It never recalls a prior sub-threshold peak.

If historical peaks are operationally indispensable, one must first apply a valid adjuster. A sufficient form is a nondecreasing function $A : [1,\infty] \to [0,\infty]$ satisfying

$$\int_1^\infty \frac{A(x)}{x^2}\, dx \le 1,$$

and then use $A(\sup_{s\le t} E_{j,s})$. The adjustment restores validity but generally reduces power.

F Proposal-Rank Budget Allocation

The spending weights determine the evidence boundary assigned to proposal rank. The following identity is an oracle calculation for the first useful proposal; it is not a claim that the default telescoping schedule is generally delay-optimal in a dynamic step-up process.

Proposition 8 (Oracle one-discovery boundary). *Let J denote the rank of the first useful proposal and suppose $P(J = j) = \pi_j$. Assume $\sum_j \pi_j = \sum_j \gamma_j = 1$ and $\gamma_j > 0$ whenever $\pi_j > 0$. Before another discovery has lowered the step-up boundary,*

$$\mathbb{E}_\pi\left[\log \frac{1}{q\gamma_J}\right] = \log\frac{1}{q} + H(\pi) + \mathrm{KL}(\pi \,\|\, \gamma). \tag{34}$$

The expectation is uniquely minimized by $\gamma = \pi$. If, in addition, the log evidence grows deterministically as $\log E_{J,\tau_J+n} = rn$ for a common $r > 0$, then replacing the oracle schedule by γ adds $\mathrm{KL}(\pi \,\|\, \gamma)/r$ to expected continuous crossing time. On an integer inspection grid the equality holds up to less than one inspection interval from rounding.

Proof. Direct expansion gives

$$\begin{aligned}\mathbb{E}_\pi\left[\log\frac{1}{q\gamma_J}\right] &= \log\frac{1}{q} + \sum_j \pi_j \log\frac{1}{\gamma_j} \\ &= \log\frac{1}{q} + H(\pi) + \sum_j \pi_j \log\frac{\pi_j}{\gamma_j}.\end{aligned}$$

The last sum is $\mathrm{KL}(\pi \,\|\, \gamma)$, which is nonnegative and vanishes exactly when $\gamma = \pi$. Under deterministic common log growth, crossing time is the log boundary divided by r, which proves the final statement. □

When proposal ranks and evidence growth rates are heterogeneous or dependent, minimizing expected hitting time becomes a different design problem. The identity nevertheless quantifies the log-boundary regret of a misspecified rank prior and explains why proposal-time economic information can be useful without allowing post-proposal outcomes to determine the budget.

G Finite-Sample Power and Certification Delay

The following bound makes the price of a late proposal explicit. It is finite-sample and allows arbitrary dependence compatible with the stated conditional alternative.

Proposition 9 (Certification by a fixed post-proposal horizon). *Fix candidate j. Suppose that, for every $s \ge 1$,*

$$X_{j,\tau_j+s} \in [-1,1], \qquad \mathbb{E}_P[X_{j,\tau_j+s} \mid \mathcal{F}_{\tau_j+s-1}] \ge \mu \tag{35}$$

almost surely, where $0 < \mu \le 1$. Use the constant predictable bet $\lambda = \mu/2$. Set

$$g = \frac{\mu^2}{4}, \qquad c_\mu = \log\frac{1+\mu/2}{1-\mu/2},$$

and let $d = |D_{\tau_j-1}|$, with $D_{-1} = \varnothing$. Conditional on $\mathcal{F}_{\tau_j}$, define

$$L_j^{\mathrm{floor}}(d) = \log\frac{1}{q} + \left[\log\frac{1}{\gamma_j(d+1)}\right]_+. \tag{36}$$

Suppose candidate j is eligible and the gate is inspected at $t = \tau_j + n$. For any $\delta \in (0,1)$, it is promoted no later than that inspection with conditional probability at least $1-\delta$ whenever

$$ng - c_\mu\sqrt{\frac{n}{2}\log\frac{1}{\delta}} \ge L_j^{\mathrm{floor}}(d). \tag{37}$$

Equivalently, it suffices that

$$n \ge \left[\frac{c_\mu\sqrt{\frac{1}{2}\log(1/\delta)} + \sqrt{\frac{1}{2}c_\mu^2\log(1/\delta) + 4gL_j^{\mathrm{floor}}(d)}}{2g}\right]^2. \tag{38}$$

Proof. For $|u| \le 1/2$, $\log(1+u) \ge u - u^2$. Put $Y_{j,t} = \log(1+\lambda X_{j,t})$. Because $\lambda = \mu/2 \le 1/2$, Eq. (35) implies

$$\begin{aligned}\mathbb{E}_P[Y_{j,t} \mid \mathcal{F}_{t-1}] &\ge \lambda\mathbb{E}_P[X_{j,t} \mid \mathcal{F}_{t-1}] - \lambda^2\mathbb{E}_P[X_{j,t}^2 \mid \mathcal{F}_{t-1}] \\ &\ge \lambda\mu - \lambda^2 = \frac{\mu^2}{4} = g.\end{aligned} \tag{39}$$

Each log increment belongs to $[\log(1-\lambda), \log(1+\lambda)]$, an interval of length c_μ. Let

$$Z_s = Y_{j,\tau_j+s} - \mathbb{E}_P[Y_{j,\tau_j+s} \mid \mathcal{F}_{\tau_j+s-1}].$$

These are martingale differences in the shifted filtration. The conditional Hoeffding lemma gives $\mathbb{E}_P[e^{-\eta Z_s} \mid \mathcal{F}_{\tau_j+s-1}] \le e^{\eta^2 c_\mu^2/8}$. Multiplication over s, Markov's inequality, and optimization over $\eta > 0$ yield

$$P\left(\sum_{s=1}^{n} Z_s \le -a \,\middle|\, \mathcal{F}_{\tau_j}\right) \le \exp\left(-\frac{2a^2}{nc_\mu^2}\right).$$

Combining this inequality with Eq. (39) and setting $a = c_\mu\sqrt{\frac{n}{2}\log(1/\delta)}$ gives

$$P\left(\log E_{j,\tau_j+n} < ng - c_\mu\sqrt{\frac{n}{2}\log\frac{1}{\delta}} \,\middle|\, \mathcal{F}_{\tau_j}\right) \le \delta. \tag{40}$$

On the complementary event, Eq. (37) yields

$$E_{j,\tau_j+n} \ge \max\left\{\frac{1}{q}, \frac{1}{q\gamma_j(d+1)}\right\}.$$

If j was promoted earlier, there is nothing to prove. Otherwise let $m = |D_{\tau_j+n-1}|$. Nestedness implies $m \ge d$, and every one of the m old discoveries remains above its threshold when the proposed set size grows from m to $m+1$. Moreover,

$$E_{j,\tau_j+n} \ge \max\left\{\frac{1}{q}, \frac{1}{q\gamma_j(d+1)}\right\} \ge \max\left\{\frac{1}{q}, \frac{1}{q\gamma_j(m+1)}\right\}.$$

Thus $k = m+1$ is feasible in the maximal step-up rule, and j is promoted at the current inspection. This proves the probability statement.

To solve Eq. (37), put $y = \sqrt{n}$ and $a = c_\mu\sqrt{\frac{1}{2}\log(1/\delta)}$. The required inequality is $gy^2 - ay - L_j^{\text{floor}}(d) \ge 0$. Its nonnegative root is

$$y = \frac{a + \sqrt{a^2 + 4gL_j^{\text{floor}}(d)}}{2g},$$

which gives Eq. (38). ☐

For $\gamma_j = 1/\{j(j+1)\}$,

$$L_j^{\text{floor}}(d) = \log\frac{1}{q} + \left[\log\frac{j(j+1)}{d+1}\right]_+.$$

Since $c_\mu = O(\mu)$ and $g = \mu^2/4$, the sufficient delay is

$$O\left(\frac{\log j + \log(1/q) + \log(1/\delta)}{\mu^2}\right). \tag{41}$$

The penalty for proposal rank is logarithmic, while weak conditional drift incurs the usual inverse-square delay. A mixture over betting fractions or a predictable plug-in bet avoids knowing μ in advance, at a logarithmic power cost, provided each component is a global e-process.

H From Certified Discoveries to Capital

The false-discovery certificate concerns a set of hypotheses. It translates directly into a capital statement only under an allocation rule that preserves the relevant self-consistency.

Corollary 10 (Equal weights). *Suppose every member of D_t receives the same gross sleeve weight. Set the null-capital fraction to zero when $D_t = \varnothing$. Then*

$$\mathbb{E}_P\left[\sup_{t\ge 0}\frac{\text{gross capital in null sleeves at } t}{\text{gross strategy capital at } t}\right] \le q. \tag{42}$$

Proof. Under equal gross sleeve weights, the displayed ratio is exactly $\sum_{j\in D_t} N_j/(|D_t| \vee 1) = \text{FDP}(D_t)$ on every sample path. Apply Theorem 6. ☐

Corollary 11 (E-budgeted allocation). *Let $a_{j,t} \ge 0$ be adapted gross-allocation shares satisfying*

$$\begin{aligned} &\sum_j a_{j,t} \le 1, \\ &a_{j,t} = 0 && \text{for } j \notin D_t, \\ &a_{j,t} \le q\gamma_j \widetilde{E}_j && \text{for } j \in D_t. \end{aligned} \tag{43}$$

Then

$$\mathbb{E}_P\left[\sup_{t\ge 0}\sum_{j=1}^{\infty} N_j a_{j,t}\right] \le q. \tag{44}$$

Proof. For every t, pathwise,

$$\sum_j N_j a_{j,t} \le q\sum_{j\in D_t}\gamma_j N_j \widetilde{E}_j \le q\sum_{j=1}^{\infty}\gamma_j N_j \overline{E}_j.$$

The right-hand side is independent of t. Take the supremum and repeat the Tonelli and conditional-e-validity argument in the proof of Theorem 6. ☐

For example, a portfolio optimizer with gross capacity G may solve

$$\max_{w\in\mathcal{W}}\left\{\widehat{\mu}_t^\top w - \frac{\rho}{2}w^\top\widehat{\Sigma}_t w - c\|w - w_{t-1}\|_1\right\}$$

subject to

$$\operatorname{supp}(w) \subseteq D_t, \qquad \|w\|_1 \le G, \qquad \frac{|w_j|}{G} \le q\gamma_j\widetilde{E}_j. \tag{45}$$

With $a_{j,t} = |w_{j,t}|/G$, Corollary 11 controls false gross exposure as a fraction of capacity. It controls the fraction of *deployed* gross capital only when $\|w_t\|_1 = G$. With variable deployment, that interpretation requires defining $a_{j,t} = |w_{j,t}|/\|w_t\|_1$ and enforcing Eq. (43) for those normalized shares.

Count-FDR alone does not justify arbitrary filtering after certification. A downstream optimizer can place most capital in a false member of an otherwise valid discovery set. Nor does an e-value measure effect size. The certificate does not, by itself, control drawdown, CVaR, turnover, realized Sharpe ratio, or estimation error in expected returns and covariances. Those quantities require a separate risk model and, where a statistical guarantee is desired, confidence sequences or other time-uniform estimators. SAFE-ALPHA's role is narrower and auditable: it limits false admission to the strategy set, and it limits false gross exposure only when the allocation layer preserves the stated e-budget.

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